

Real-Time Diagnostics of Hydraulic Fracture Propagation Using Continuous Wavelet Transform of Transient Pressure Fluctuation

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Summary

Hydraulic fracturing diagnostics are essential for the efficient stimulation of unconventional reservoirs. Treating pressure is an accessible field signal and provides a low-cost and nonintrusive basis for characterizing hydraulic fracture propagation. However, conventional time-domain analysis often fails to capture transient fluctuations associated with fracture propagation because of the interference from wellbore resonance, operational adjustments, and background noise. More valuable pressure fluctuation distributes across the various frequency components. In this paper, we propose a real-time diagnostic approach using the continuous wavelet transform (CWT) to extract rate-corrected transient pressure fluctuations intrinsic from the stress release and accumulation of hydraulic fractures propagation. It enables the dynamic classification of fracture propagation patterns, including primary fracture growth, complex fracture development, discontinuities intersection, and fracture interruption. CWT in particular is used to decompose the treating pressure signal, and the fracture-induced fluctuations are characterized as broadband impulsive responses in the time-frequency energy spectrum. Wavelet coherence analysis between pressure and slurry rate is introduced to identify time-frequency regions minimally disturbed by variable rates. Results indicated that the 0.3–0.5-Hz frequency components are identified as an effective diagnostic band for data sampling of 1 Hz. Pressure fluctuation energy in the 0.4–0.5-Hz sub-band serves as a surrogate for instantaneous stress release of hydraulic fracture propagation and reflects the efficiency with which the hydraulic energy is converted into new fracture surfaces. The pressure fluctuation attenuation coefficient is defined from the pressure energy distribution contrast between high- and low-frequency components. Increased hydraulic-energy dissipation along the fracture path and hindered stress reaccumulation produce frequency variation within the pressure fluctuation period, leading to a stronger transfer of pressure energy toward the 0.3–0.4-Hz sub-band. Based on these two quantified features, rule-based fuzzy inference is then used to classify the fracture propagation patterns into four categories on a 5-second window basis. Supporting evidence for the proposed method was provided by independent microseismic and distributed fiber-optic measurements. The fracturing stages dominated by complex fracture development are further supported by higher interstage tracer production. In addition, locally prolonged elevated energy near 0.35–0.4 Hz in the pressure time-frequency spectrum is identified as a potential indicator of enhanced fracture complexity associated with large-scale discontinuities activation and fracture branching. The proposed technique provides an automated, quantitative connection between treating pressure fluctuations and fracture propagation dynamics, offering a practical real-time alternative for the evaluation of hydraulic fracturing performance.

Introduction

Hydraulic fracturing diagnostics are essential for the efficient stimulation of unconventional reservoirs. Understanding fracture propagation behavior is critical for evaluating treatment performance and guiding real-time optimization of pumping parameters. As the routinely recorded field signal, treating pressure provides an economical and nonintrusive solution for diagnosing fracture propagation dynamics. It reflects the integrated response to treatment strategy, resonances in the fluid column from wellbore or fracture deformation, propagation of fluid connected fractures, and background noise (MacKay et al. 2018; Wei et al. 2025). Accurate characterization of pressure evolution associated with distinct fracture propagation patterns is fundamental to pressure-based diagnostics.

Nolte and Smith (1981) pioneered pressure-based diagnostics through log-log interpretation of net pressure vs. time. Subsequent studies extended this framework to pressure-trend classification (stable, fluctuation, rise, and drop) for identifying fracture propagation patterns such as normal propagation, fracture-height growth, and proppant screenout (Chen et al. 2020). However, the Nolte-Smith method generally relies on assumptions including continuous fracture propagation, fixed pumping rate, or prior knowledge of closure pressure, which may not hold during large-scale volume fracturing. Besides, compression of pressure data imposed by the log-log plot makes real-time interpretation hard. The Nolte-Smith framework was further extended by introducing intermittent fracture propagation with alternating growth and ballooning periods, which led to the moving reference point (MRP) method for real-time diagnosis of fracturing events (Soliman et al. 2014; Pirayesh et al. 2015). Although MRP does not require prior knowledge of closure pressure and presents results in Cartesian coordinates, it still depends on a relatively complex mathematical formulation for determining the moving reference exponent. Numerical simulations based on Biot's theory indicate that hydraulic fracture propagation is inherently intermittent. Fracture opening releases stress and causes a rapid pressure drop, while subsequent fluid filling drives stress reaccumulation and pressure recovery until the next rupture (Tzschichholz and Herrmann 1995; Milanese et al. 2016; Cao et al. 2018). As a result, fracture growth leaves transient

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pressure fluctuations superimposed on the global pressure trend. Transient pressure fluctuations have also been observed in laboratory true-triaxial hydraulic fracturing tests. Bian et al. (2016) related fluctuation frequency and amplitude on the treatment curve to fracture complexity and far-field fracture geometry. Abdelaziz and Grasselli (2024) observed the difference in pressure fluctuation between tensile and shear fractures, which provided key physical evidence for the characteristics of pressure fluctuation. Tensile fracture exhibits a sharp pressure drop with quick recovery, while shear fracture shows a slower pressure recovery. These observations indicate that pressure fluctuations are informative indicators of complex fracture evolution. Owing to formation heterogeneity and varying fracture types, such fluctuations appear not as continuous water-hammer oscillations, but as frequency-varying transient responses.

Because fracture-induced pressure fluctuations are nonstationary and frequency-varying, time-domain analysis alone is often insufficient. Wavelet transform provides localized resolution in both time and frequency, enabling effective capture of abrupt singularities with less temporal smearing than Fourier-based global transforms (Mallat and Hwang 2002; Daubechies 1990; Addison 2005). It is therefore well suited for preserving rupture-related energy bursts and identifying frequency transitions in fracturing pressure signals. Accordingly, wavelet transform has recently been applied to pressure-based fracture diagnostics without requiring simplifying assumptions or sophisticated models (Li et al. 2022a; Sheng et al. 2022; Hu et al. 2025). Wavelet transform mainly includes discrete wavelet transform (DWT) and CWT. DWT has been used in pumping pressure and diagnostic fracture injection test (DFIT) analysis to identify events such as leakoff, ballooning, and fracture closure, and to reduce rate-related interference through pressure-rate wavelet-energy comparison (Unal et al. 2018, 2019b; Hu et al. 2022). While DWT offers efficient computation, its fixed decomposition scales and substantial downsampling inherently limit frequency resolution in high-frequency components. By contrast, CWT provides superior frequency resolution through continuous decomposition scales, making it better suited for capturing the evolution of transient responses in nonstationary pressure signals (Arts and van den Broek 2022). Although the computational cost of CWT is more intensive, CWT maintains acceptable processing efficiency for field data with one-second interval while preserving high-frequency components resolution. Gabry et al. (2023a, 2023b) applied CWT to DFIT analysis and identified fracture closure from minimum stabilized signal energy during pressure falloff, with support from field-scale strain measurements. Wei et al. (2024) applied CWT to post-fracturing pressure falloff data to identify the nearly pure leakoff and invert fracture area, and the CWT-derived estimates correlated more strongly with stage production. Gabry et al. (2024a, 2024b) also generated normalized CWT scalograms of pumping pressure to interpret 3D fracture-propagation modes in comparison with microseismic observations. In addition, Zhang et al. (2025) applied CWT-based time-frequency features of treating pressure to identify multifracture propagation modes and evaluate diverting effectiveness, capturing the transition from frequent pressure fluctuations under fracture competition to gradual stabilization as primary fractures became established.

Despite these advances, previous studies mainly focused on the propagation and closure behavior of primary hydraulic fractures, with limited attention to interactions between hydraulic fractures and discontinuities such as natural fractures, bedding planes, and interlayers. Fundamentally, fracture propagation emerges from competition between tensile and shear failure mechanisms under hydraulic loading. Primary hydraulic fractures are typically tensile-dominated, whereas interaction with pre-existing discontinuities promotes shear-dominated failure and complex fracture-network development (Warpinski and Teufel 1987; Wu et al. 2022; Li et al. 2025; Xu et al. 2026). These competing mechanisms also affect transient pressure fluctuations: Rapid pressure recovery indicates relatively simple fracture paths with efficient stress redistribution and limited hydraulic-energy dissipation, whereas slower recovery reflects increased dissipation and hindered stress reaccumulation associated with more tortuous propagation and complex discontinuity interaction. Therefore, establishing a quantitative link between fracture-propagation stress states and CWT signatures of transient pressure fluctuations is essential for expanding the diagnostics of fracture-discontinuity interaction.

To address the above issues, we develop a real-time diagnostic method that links transient pressure fluctuations to stress release and accumulation under different interaction scenarios between hydraulic fractures and natural discontinuities. Starting from routinely acquired treating pressure under standard 1-Hz sampling, rate-decoupled transient fluctuations are extracted in the time-frequency domain using CWT. Two key features are defined to quantify the transient pressure fluctuations intrinsic from the stress release and accumulation of hydraulic fractures propagation and are further input into a fuzzy inference system for dynamic classification of fracture propagation patterns (Fig. 1). The fracture-related physical processes underlying the proposed method are supported by independent field measurements, including microseismic monitoring, distributed fiber-optic observations, and stage-level tracer production data. Overall, it would provide a low-cost, noninvasive tool for evaluating fracture propagation events and complexity during hydraulic fracturing.

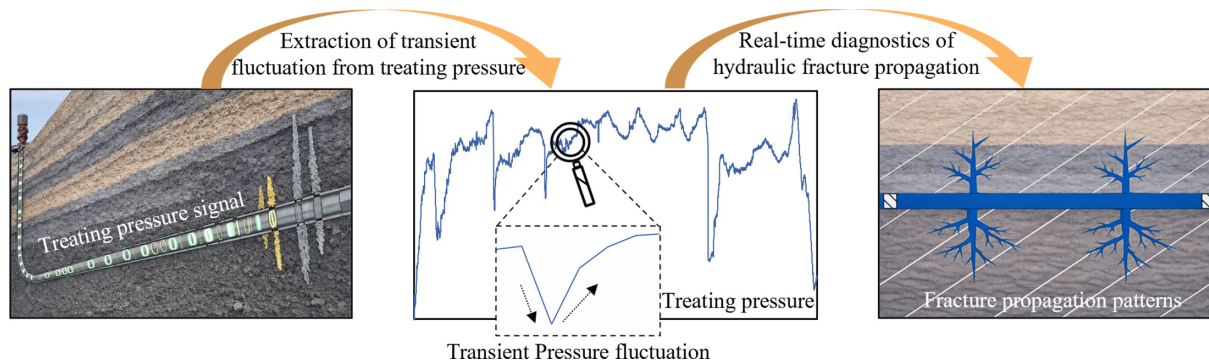


Fig. 1—Schematic framework of fracture diagnostics based on transient pressure analysis.

Methodology

Workflow of the Proposed Method. A transient-pressure analysis workflow is developed for diagnosing fracture propagation patterns during hydraulic fracturing. It consists of three main steps. First, the CWT is applied to decompose treating pressure in the time-frequency domain and extract rate-decoupled transient fluctuations using wavelet coherence analysis. Second, the target diagnostic band is determined and two features of pressure fluctuation energy and attenuation are defined to quantify the transient pressure fluctuations intrinsic from the stress release and accumulation of hydraulic fractures propagation. Third, based on the two quantified

features, a rule-based fuzzy inference system is constructed to classify fracture propagation behaviors into four distinct patterns: primary fracture growth, complex fracture development, discontinuities intersection, and fracture interruption. **Fig. 2** summarizes the overall workflow.

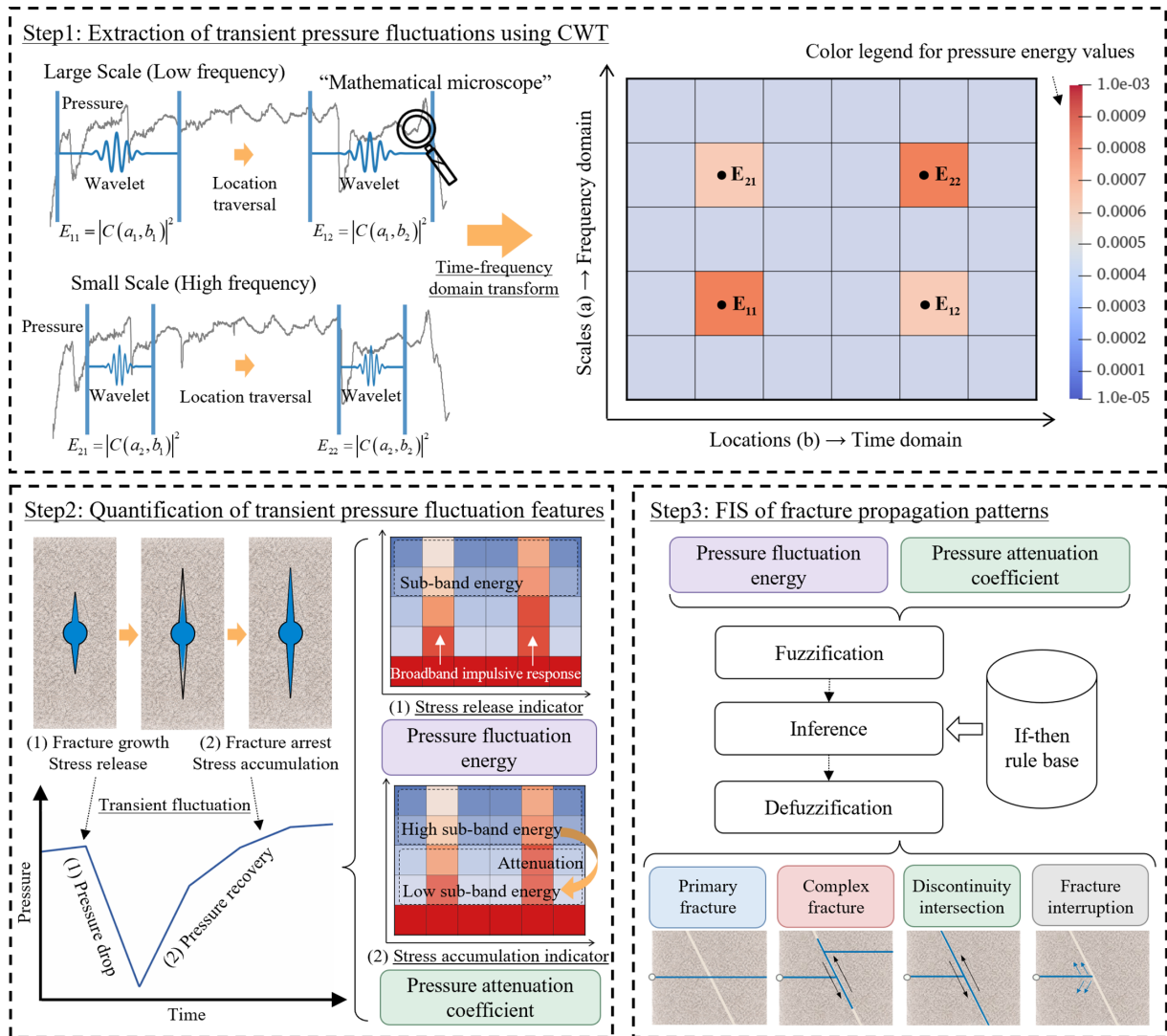


Fig. 2—Transient-pressure analysis workflow for diagnosing fracture propagation patterns. FIS = fuzzy inference system.

Extraction of Transient Pressure Fluctuations Using CWT. Transient pressure fluctuations are often subtle and short-lived, and are easily obscured in raw time-domain pressure records by operational adjustments, wellbore resonance, and background noise. Time-frequency analysis is therefore required to separate fracture-induced pressure fluctuations from interference. As a mathematical microscope for transient-signal detection, CWT enables adaptive time-frequency resolution and has been widely applied in mechanical fault diagnosis (Nikolaou and Antoniadis 2002; Yan et al. 2014). By performing scale and translation operations on a selected mother wavelet, CWT decomposes 1D signal into multiple frequency components, allowing the detailed examination of fluctuations at different frequencies. CWT of a signal $f(t)$ is defined as follows (Addison 2017):

$$C(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} f(t) \varphi\left(\frac{t-b}{a}\right) dt, \quad (1)$$

where $f(t)$ is the time-series signal, and φ is the mother wavelet that becomes φ^* when employing a complex wavelet. The scale parameter a controls the frequency of the wavelet by compressing or stretching the mother wavelet. A smaller scale corresponds to a higher frequency, whereas a larger scale corresponds to a lower frequency. The location parameter b determines the time localization of the wavelet. The wavelet coefficients $C(a, b)$ are obtained through the convolution of the signal with the wavelet across various scales and locations, serving as an indicator of their local similarity. The complex Morlet wavelet provides smooth energy continuity and high-frequency resolution due to its Gaussian-window form. Its complex representation enables simultaneous extraction of amplitude and phase, making it well suited for identifying transient high-frequency responses associated with fracture behavior. Practical applications show that the complex Morlet wavelet with a bandwidth of 1.5 and a center frequency of 1 Hz is effective for identifying fracture propagation and closure events, as well as analyzing water-hammer energy attenuation under a 1-Hz sampling rate (Gabry et al. 2024b, 2025). The complex Morlet wavelet function is defined as

$$\varphi^* = \frac{1}{\sqrt{\pi B}} e^{\frac{t^2}{B}} e^{i2\pi f_c t}, \quad (2)$$

where B is the bandwidth, and f_c is the center frequency (Hz). Because the treating pressure is usually acquired at a sampling rate of 1 Hz for most field applications, the Nyquist sampling theory indicates that the effective frequency components are concentrated in the 0–0.5-Hz band (Ricci et al. 2024). Accordingly, linear frequency within the range of 0–0.5 Hz is used, and the corresponding scales are mapped using Eq. 3 (Unal et al. 2019a):

$$a = \frac{f_c}{T_s f_a}, \quad (3)$$

where T_s is the sampling period (seconds), and f_a is the frequency corresponding to scale a . A total of 256 scales are established to enhance the resolution in the high-frequency components while maintaining adequate coverage of the low-frequency components. Signal energy is a key attribute that characterizes the strength and variation of the signal (Unal et al. 2019a). CWT can effectively decompose and preserve the energy distribution of pressure fluctuation across different frequency scales, where the energy is calculated from the squared modulus of the wavelet coefficients.

$$E(a, b) = |C(a, b)|^2. \quad (4)$$

Wavelet coherence provides a powerful time-frequency tool for quantifying the coupling influence of external signals on the target signal. Based on CWT, it measures localized correlation and phase consistency between two time series and is particularly effective for analyzing nonstationary signals (Peng et al. 2020). The wavelet coherence analysis consists of three main steps. First, CWT is applied separately to the treating pressures and slurry rates to obtain their time-frequency energy spectrum $E_x(a, b)$ and $E_y(a, b)$, respectively. Subsequently, the time-frequency cross spectrum $E_{xy}(a, b)$ is estimated by Eq. 5:

$$E_{xy}(a, b) = C_x(a, b) C_y^*(a, b), \quad (5)$$

where $C_x(a, b)$ and $C_y(a, b)$ are the wavelet coefficients of the pressure and rate signals across various scales and locations, and $*$ denotes the complex conjugate. To reduce the influence of random noises, Gaussian smoothing is applied to both the individual and cross spectra. Finally, the wavelet coherence coefficient is calculated as

$$R^2(a, b) = \frac{S[E_{xy}(a, b)]}{S[E_x(a, b)] S[E_y(a, b)]}, \quad (6)$$

where $S[\cdot]$ denotes the Gaussian smoothing operator. It can be defined as

$$S(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{t^2}{2\sigma^2}}, \quad (7)$$

where t is the time (seconds), and σ is the standard deviation that controls the width of the Gaussian kernel and the degree of smoothing. The wavelet coherence coefficients vary within the interval $[0, 1]$, where a value close to 1 indicates a strong correlation.

Quantification of Transient Pressure Fluctuation Features. Building on the time-frequency decomposition, this section converts the CWT-resolved transient fluctuations into quantitative metrics. During fracture propagation, intermittent stress release and reaccumulation generate transient pressure-drop and pressure-recovery fluctuations superimposed on the global pressure trend. Unlike continuous sinusoidal oscillations, these transient fluctuations exhibit asymmetric half-cycles and corresponding frequency variation, exciting broadband impulsive responses in the time-frequency domain characterized by rapid low-to-high frequency energy transfer and nonuniform energy distribution across frequencies. On this basis, two quantitative features—pressure fluctuation energy and attenuation coefficient—are defined to characterize the stress release and reaccumulation intrinsic to hydraulic fractures propagation. The pressure fluctuation energy is calculated within a diagnostic frequency sub-band by integrating along the frequency axis:

$$E_{band}(\bar{a}, b) = \frac{1}{a_1 - a_2} \int_{a_2}^{a_1} E(a, b) da, \quad (8)$$

where $E_{band}(\bar{a}, b)$ denotes the mean pressure energy in the target sub-band, and a_1 and a_2 are the scales corresponding to the lower and upper cutoff frequencies, respectively. Consequently, pressure fluctuation energy provides a quantitative proxy for how efficiently the hydraulic energy is converted into new fracture surfaces within the wellbore/formation system. Higher conversion efficiency corresponds to stronger instantaneous stress release from new fracture surfaces, which, in turn, tends to excite higher pressure fluctuation energy. Furthermore, the pressure peak energy is identified to characterize the temporal distribution of prominent fracture-propagation events during the treatment. The detection of peak energy combines both a local threshold and a global threshold. The local threshold baseline δ_{local} is constructed using the exponential moving average and the corresponding standard deviation:

$$\begin{cases} \delta_{local} = \mu_i + 2\sigma_i \\ \mu_i = \alpha E_i + (1 - \alpha) \mu_{i-1} \\ \sigma_i = \sqrt{\alpha (E_i - \mu_i)^2 + (1 - \alpha) \sigma_{i-1}^2} \end{cases}, \quad (9)$$

where i is the time index (seconds); δ_{local} is the threshold at the i th step; μ_i and σ_i are the sample mean and standard deviation at the i th step; α is the smoothing coefficient, set to 0.05; E_i is the pressure energy at the i th step; and μ_{i-1} and σ_{i-1} are the sample mean and standard deviation at the previous timestep. The local threshold adapts to nonstationary fluctuations and highlights the short-term significance

of pressure fluctuation energy. In contrast, the global threshold provides a stable baseline for cross-comparison among different lithologies and reservoirs, ensuring consistent peak-discrimination criteria to suppress background noise. Specifically, the global threshold baseline δ_{global} can be statistically determined as the minimum level of the pressure energy distribution across different lithologies and formations. The minimum level is set to the order of 25% quantile according to statistical characteristics. As a result, the peak energy detection algorithm identifies local maxima of the pressure fluctuation energy that satisfy the following conditions.

$$\begin{cases} E_{i-1} < E_i > E_{i+1} \\ E_i \geq \max(\delta_{local}, \delta_{global}) \end{cases} \quad (10)$$

A pressure-fluctuation filter is then applied to remove false energy peaks. For each detected peak energy, the presence of a negative pressure trough is checked within a ± 5 -second window. If no negative trough is found, the peak is treated as a pressure discontinuity inconsistent with the fracture-induced fluctuation, and the corresponding pressure energy is set to zero. This ensures that only reliable pressure energy peaks are retained to represent actual fracture propagation events.

In addition, the pressure fluctuation attenuation coefficient is defined from the pressure energy distribution contrast between high- and low-frequency sub-bands. To compress the amplitude difference in pressure energy and enhance the contrast of different frequency components, the pressure time-frequency energy spectrum is logarithmically transformed:

$$E_{\log}(a, b) = \log_2 [E(a, b)] \quad (11)$$

The log-transformed pressure energy within each sub-band is then calculated using Eq. 8, yielding two time series of the mean pressure energy denoted as $X = [x_1, x_2, \dots, x_n]$ and $Y = [y_1, y_2, \dots, y_n]$. When the evolution trends of pressure energy between sub-bands begin to diverge, it indicates the occurrence of pressure fluctuation attenuation, with a stronger transfer of pressure energy toward the low-frequency band. A 30-second sampling window with a 5-second step size is applied to traverse the entire time series, and the Pearson correlation coefficient between X and Y is calculated within each sampling window:

$$r_{XY} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (12)$$

where r_{XY} is the Pearson correlation coefficient, the value range of which is $[-1, 1]$, n is the total length of the sampling window (seconds), x_i and y_i are the values of mean normalized pressure energy in the high- and low-frequency bands, and $\bar{x}$ and $\bar{y}$ are their respective averages. Furthermore, the correlation coefficients of each sampling window are linearly interpolated to match the original time-series length. The interpolated correlation coefficient $\hat{r}_{XY}$ is then integrated with the absolute difference in pressure energy between the two series to quantify the pressure fluctuation attenuation coefficient:

$$\begin{cases} \Delta E_{XY}^i = x_i - y_i \\ D_{XY}^i = \max\{0, \min[(1 - \hat{r}_{XY}), 1]\} \times \max(0, \Delta E_{XY}^i) \end{cases} \quad (13)$$

where ΔE_{XY}^i is the absolute difference in pressure energy, and D_{XY}^i is the pressure fluctuation attenuation coefficient at a given time. Similarly, the global threshold baseline of the pressure fluctuation attenuation coefficient across different lithologies and formations can be determined as the minimum level of statistical distribution, consistent with its order of 25% quantile. Pressure fluctuation attenuation is defined as a quantitative proxy for the degree of hydraulic-energy dissipation along the fracture path within the wellbore/formation system. When the energy input is consumed by additional dissipation after the creation of new fracture surfaces, stress reaccumulation becomes slower, resulting in increased attenuation. Such dissipation is commonly associated with the influence of discontinuities, such as natural fractures, bedding planes, and interlayers.

Based on the two quantitative features, transient pressure fluctuations can be classified into four representative behaviors that interpret the tendency of hydraulic energy allocation within the wellbore/formation system and correspond to four common fracture propagation patterns (**Fig. 3**). (i) Primary fracture growth: Most hydraulic energy is efficiently converted into new fracture surfaces, with limited additional dissipation. Fractures propagate actively through the intact matrix under tensile opening, resulting in high pressure fluctuation energy and low attenuation. (ii) Complex fracture development: Hydraulic energy is partly converted into new fracture surfaces, but additional dissipation increases as fractures deviate along weak discontinuities, generating branches and tortuous paths. In this case, the competition of tensile and shear failure leads to elevated energy and attenuation. (iii) Discontinuities intersection: Hydraulic energy is preferentially captured by pre-existing discontinuities, making new fracture-surface creation difficult. Increased shear failure and hydraulic energy dissipation retard stress release and reaccumulation, leading to reduced energy and higher attenuation. (iv) Fracture interruption: Hydraulic energy is further dissipated within nonfracture spaces of the wellbore/formation system, hindering hydraulic fracture propagation. Under increased leakoff and limited fracture volume, stress variations are too weak to sustain transient pressure fluctuations, so both energy and attenuation remain at low levels. These categories reflect hydraulic-energy allocation within the wellbore/formation system rather than unique fracture geometries. Accordingly, the represented scenarios are broader than the schematic interactions in **Fig. 3**: Fracture branching, redirection, and larger-scale fracture network may all be categorized as “Complex fracture development,” whereas nongeological dissipation, such as interstage communication, may also be interpreted as “Discontinuities intersection.”

Fuzzy Inference System of Fracture Propagation Patterns. With the two pressure-fluctuation features quantified, the remaining task is to translate these continuous metrics into interpretable fracture propagation patterns. Because field signals are noisy and the boundaries between fracture behaviors are not strictly separable, a soft-decision framework is preferred over hard thresholds. Fuzzy logic accommodates such uncertainty by representing inputs with membership functions and linguistic if-then rules. In this study, a Mamdani-type fuzzy inference system (Mamdani and Assilian 1975) is adopted for its transparency and ease of incorporating expert knowledge. The workflow includes fuzzification, rule-based inference, and defuzzification to obtain the final fracture-pattern label.

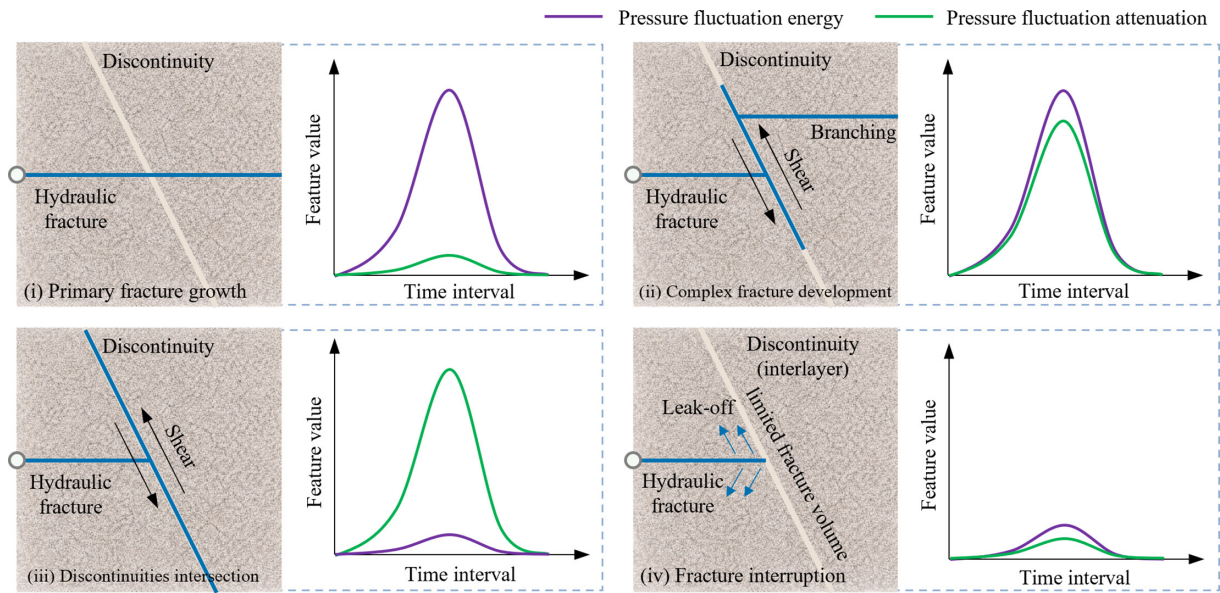


Fig. 3—Four dominant fracture propagation patterns classified by pressure fluctuation energy and attenuation coefficient.

For real-time diagnosis, a rule-based fuzzy classification algorithm is established using pressure fluctuation energy E_i and attenuation coefficient D_{XY}^i as key indicators. The 1-Hz time series is segmented into nonoverlapping 5-second windows, which approximately encompass the typical 2–5 seconds duration of individual transient pressure fluctuations. Within each window, the median and median absolute deviation (MAD) are calculated for both E_i and D_{XY}^i to serve as robust feature inputs:

$$\begin{cases} \hat{X} = \text{median}(X_i) \\ MAD_X = 1.4826 \times \text{median}(|X_i - \hat{X}|) \end{cases}, \quad (14)$$

where $\hat{X}$ is the median of the sampled data X_i , and MAD_X is the median of the absolute deviations from the median, which is normalized using the factor 1.4826 under Gaussian assumptions. The ratio of outliers is defined as values exceeding k times MAD from the median:

$$\begin{cases} R_E = \frac{1}{n} \sum_{i=1}^n \left[|E_i - \hat{E}| k \cdot MAD_E \right] \\ R_D = \frac{1}{n} \sum_{i=1}^n \left[|D_i - \hat{D}| k \cdot MAD_D \right] \end{cases}, \quad (15)$$

where $\hat{E}$ and $\hat{D}$ are the median within the analysis window, and k is the MAD multiplier controlling outlier sensitivity; it may be adjusted according to the temporal sparsity of transient pressure fluctuations, with sparser and more temporally concentrated responses generally favoring a smaller k . To ensure the consistency and significance of fracture-induced pressure fluctuation features under specific lithological and reservoir conditions, the 25th-percentile levels of the pressure energy and attenuation coefficient distributions are adopted as global thresholds. If the window mean does not exceed the corresponding threshold, the outlier ratio is set to zero to suppress background noise. The fuzzy system uses the ratios of pressure-energy and attenuation-coefficient outliers (R_E and R_D) as inputs, each divided into two LOW and HIGH by trapezoidal membership functions for computational efficiency and direct visualization of threshold-based decision regions in engineering diagnostics (Artun and Kulga 2020). The fuzzy inference result is discretized into four dominant fracture propagation patterns according to the rules listed in **Table 1**. Centroid defuzzification converts the fuzzy inference result into a crisp value from 0 to 4, which is rounded to the nearest integer to identify the dominant pattern for each window.

Rules	Ratio of Pressure Energy Outliers, R_E	Ratio of Attenuation Coefficient, R_D	Categories
1	LOW	LOW	Fracture interruption
2	HIGH	LOW	Primary fracture growth
3	LOW	HIGH	Discontinuities intersection
4	HIGH	HIGH	Complex fracture development

Table 1—Rules to characterize fracture propagation pattern.

Fig. 4 illustrates a working example of the rule-based fuzzy inference system within a single 5-second analysis window. Each subfigure shows the trapezoidal membership function of an input or output variable, with variable value on the x-axis and membership degree on the y-axis, and different colors denoting different fuzzy classes. For the example inputs $R_E = 0.35$ and $R_D = 0.42$, fuzzification assigns $R_E = 0.35$ to both “LOW” and “HIGH” with membership degrees determined by the corresponding trapezoidal functions, whereas $R_D = 0.42$ belongs only to “HIGH” with a membership degree of 1. This activates two rules: (1) HIGH R_E and HIGH R_D correspond to “Complex

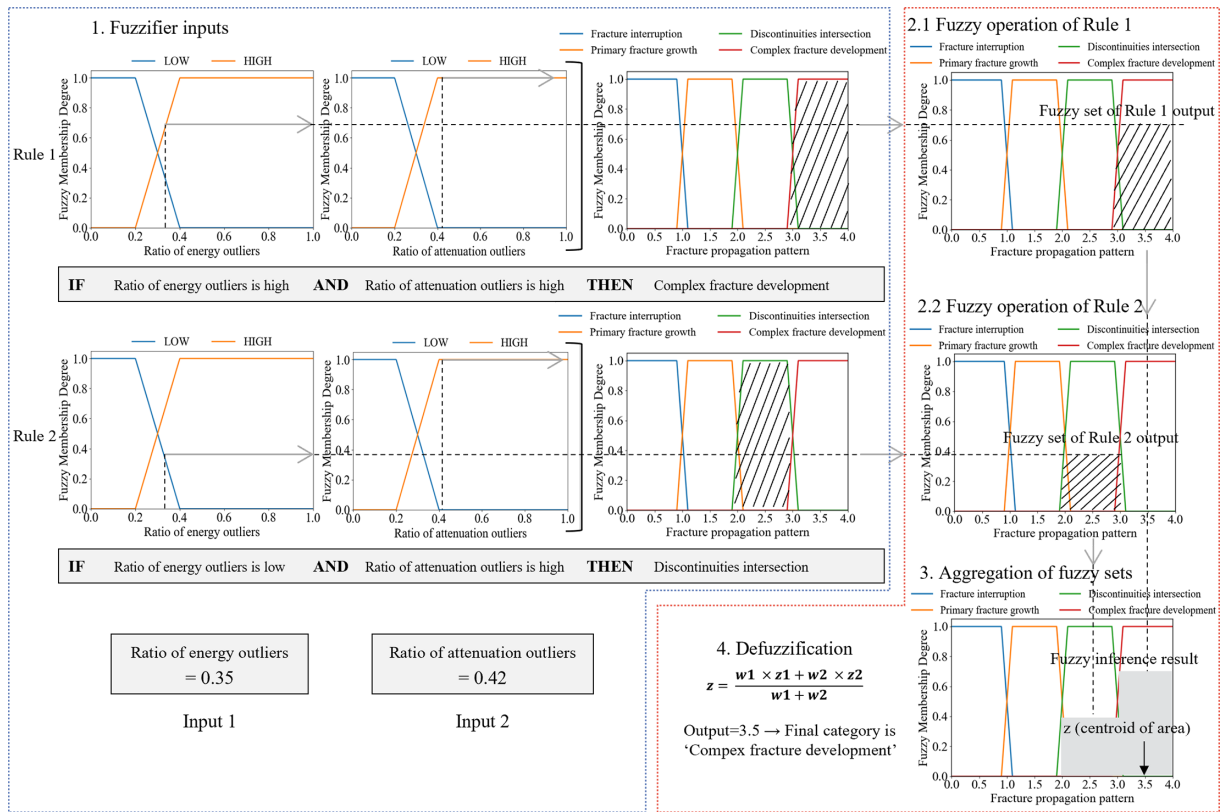


Fig. 4—Aggregation of rules using the graphical Mamdani inference method and center of gravity calculation.

fracture development,” and (2) LOW R_E and HIGH R_D correspond to “Discontinuities intersection.” Each of them represents the possible fracture pattern implied by the current inputs. Using the minimum operator, the corresponding candidate output fuzzy sets are generated, shown by the shaded hatched regions in Fig. 4. These outputs are then aggregated into the fuzzy inference result by the gray region in the lower-right panel. Centroid defuzzification finally gives an output value of 3.5, corresponding to the final classification of “Complex fracture development.”

Applying the fuzzy inference system to each analysis window yields a sequence of fracture-pattern labels. Each 5-second window characterizes the local tendency of hydraulic energy allocation within the wellbore/formation system. For engineering interpretation, these labels are more meaningfully aggregated over a longer interval, and the proportion of each category can be treated as an empirical indicator of the relative tendency of different fracture-propagation behaviors.

Results

Real-Time Diagnostics of Fracture Propagation Patterns. Pressure Time-Frequency Energy Spectrum Based on CWT. First, CWT is applied to the treating pressure signal to obtain the time-frequency distribution of pressure energy. Fig. 5 shows an example based on field data from a representative hydraulic fracturing stage in the Montney Basin. Higher amplitudes in the time-frequency energy spectrum indicate stronger pressure fluctuations at the corresponding time and frequency. Transient pressure fluctuations appear as broadband impulsive responses, characterized by a rapid low-to-high frequency energy transfer and nonuniform energy distribution across frequencies (e.g., around 2,000s and 3,010s). Although such responses are intrinsically broadband and do not theoretically require a unique diagnostic band, practical pumping adjustments can introduce rate-coupled interference into part of the spectrum. It is therefore necessary to separate rate-coupled components and identify the band most representative of fracture propagation.

The original time-frequency energy spectrum is corrected for slurry rate effects using the wavelet coherence analysis. The wavelet coherence spectrum quantifies the time-frequency correlation between pressure and rate. High-coherence regions where coefficients approach unity correspond to pressure fluctuations dominated by rate adjustments. Fig. 6a highlights significant coherence regions ($R^2(a, b) > 0.8$, gray contour), which are predominantly concentrated below 0.3 Hz. Phase difference θ between the two signals is estimated to characterize the positive or negative coherence. θ is indicated by arrow direction, where right-pointing arrows correspond to zero, counterclockwise rotation toward the left indicates zero to π , and clockwise rotation back toward the right indicates zero to $-\pi$. In the significant coherence regions, arrows predominantly point to the right, indicating $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. This confirms positive coherence and approximately in-phase variation between pressure and rate, with slight phase deviations.

In the 0.3–0.5-Hz band, wavelet coherence is insignificant, suggesting weak pressure/rate coupling. This band is therefore selected as the effective diagnostic window for 1-Hz sampled data. A localized time-domain correction is then applied by averaging the coherence coefficients along the frequency axis within this band:

$$R^2(\bar{a}, b) = \frac{1}{a_1 - a_2} \int_{a_2}^{a_1} R^2(a, b) da, \quad (16)$$

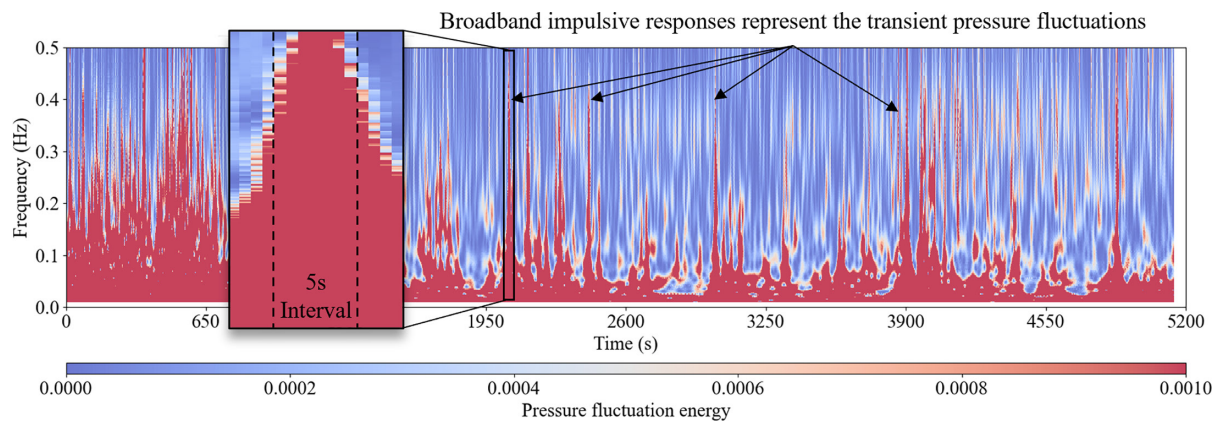


Fig. 5—Time-frequency energy spectrum derived from field treating pressure acquired from a hydraulic fracturing stage in the Montney Basin.

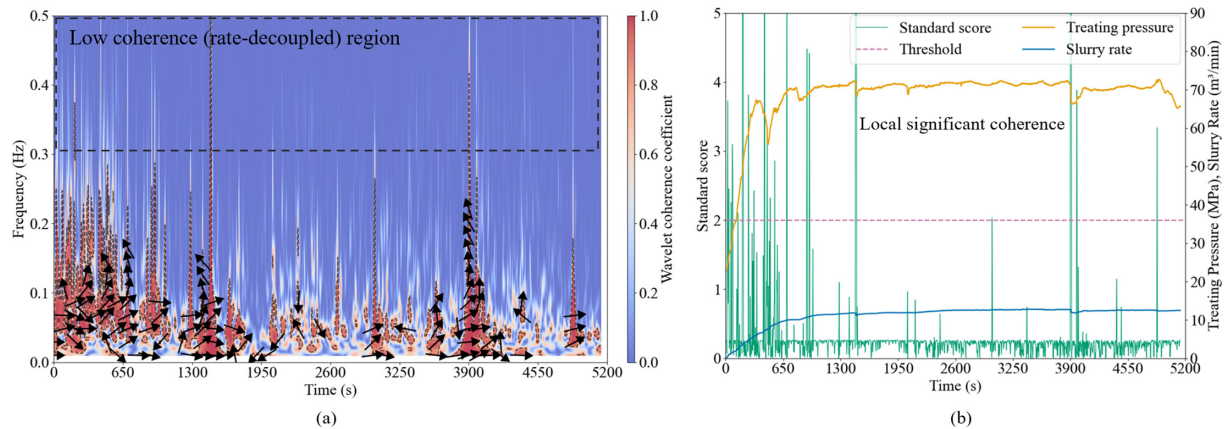


Fig. 6—(a) Wavelet coherence spectrum between treating pressures and slurry rates. (b) Localization of local significant coherence in 0.3–0.5-Hz band.

where $R^2(\bar{a}, b)$ is the mean coherence coefficient within the current subfrequency band, and a_1 and a_2 are the scales corresponding to the lower and upper cutoff frequencies, which are set to 0.3 Hz and 0.5 Hz here, respectively. Local high-coherence moments are extracted using the standard score (z-score), defined as

$$Z = \frac{|R^2(\bar{a}, b) - \bar{R}^2(\bar{a}, b)|}{S}, \quad (17)$$

where Z is the standard score, $R^2(\bar{a}, b)$ is the mean coherence coefficient at a certain moment, $\bar{R}^2_{norm}(\bar{a}, b)$ is the average value of the mean coherence coefficients at all moments, and S is the standard deviation of the mean coherence coefficients at all moments. As shown in **Fig. 6b**, a threshold of $Z > 2$ is applied to identify instants of local significant coherence. A rate-coupled interval is then defined as a ± 5 -second window surrounding each identified instant. Accordingly, the pressure energy within these intervals is corrected to zero to mitigate local rate effects.

Diagnostic Sub-Band Selection and Feature Quantification. Using wavelet coherence analysis, the 0.3–0.5-Hz range is identified as the diagnostic band for fracture-induced transient pressure fluctuations. To further define physically meaningful sub-bands within this range, reported 3D finite-discrete element method simulation results are analyzed (Cheng et al. 2026). Spearman rank correlation (de Winter et al. 2016) between sub-band pressure energy within 0–0.5 Hz and incremental fracture volume increases markedly above 0.3 Hz (**Fig. 7b**). This indicates reduced low-frequency trend interference and stronger temporal localization of broadband impulsive responses, making the 0.3–0.5-Hz band more representative of fracture propagation. Together with true-triaxial experimental observations reported by Abdelaziz and Grasselli (2024), these results further support that transient pressure fluctuations reflect the inherently unstable stress state of fracture propagation and typically exhibit asymmetric drop-recovery half-cycles (**Fig. 8**). The pressure drop, triggered by rapid fracture opening and instantaneous stress release, is short and approaches the 0.4–0.5-Hz component under 1-Hz sampling. By contrast, the pressure recovery, triggered by hydraulic-energy supplement and progressive stress reaccumulation, is longer and contributes more strongly below 0.5 Hz. The dominant failure mechanism mainly affects the recovery stage: Tensile- and shear-dominated failures show comparable pressure-drop half-cycles, but shear responses exhibit markedly longer recovery half-cycles. This produces a larger energy contrast between the 0.3–0.4-Hz and 0.4–0.5-Hz sub-bands in shear responses. Accordingly, the 0.4–0.5-Hz sub-band energy is used as a surrogate for instantaneous stress release and defines the pressure fluctuation energy in Eq. 8, while the adjacent 0.3–0.4-Hz sub-band is used as a reference to quantify pressure fluctuation attenuation. Greater energy transfer toward 0.3–0.4 Hz indicates slower stress reaccumulation and a higher likelihood of hydraulic fracture interaction with natural discontinuities.

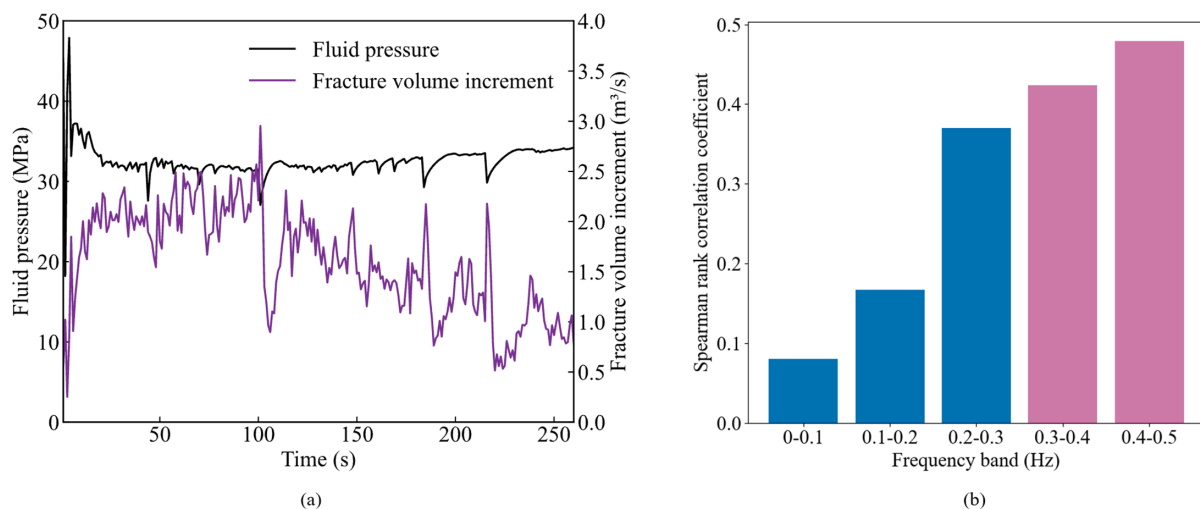


Fig. 7—(a) Fracture volume increment and the response of fluid pressure at the injection point (calculated from simulation results reported by Cheng et al. 2026). (b) Spearman rank correlation between band-limited pressure energy and fracture volume increment.

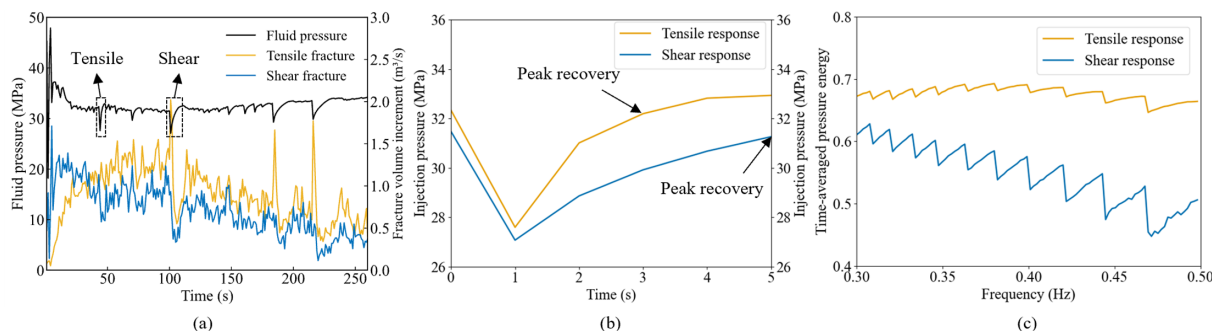


Fig. 8—(a) Tensile/shear fracture volume increment and corresponding fluid-pressure response (Calculated from simulation results reported by Cheng et al. 2026). (b) Pressure fluctuation of tensile and shear failure. (c) Frequency-domain pressure energy distribution for tensile and shear failure.

Classification Results of Fracture Propagation Patterns. Rule-based fuzzy inference is used to classify fracture propagation patterns into four categories based on these two features. Their global thresholds are statistically determined for different lithologies and reservoirs using the field data sets summarized in Table 2. As shown in Fig. 9, both pressure fluctuation energy and attenuation coefficient in shale reservoirs are approximately one order of magnitude higher than those in sandstone reservoirs, mainly because shale is more brittle and commonly contains bedding discontinuities that promote stronger stress release and more tortuous fracture propagation (Hou et al. 2016; Li et al. 2023). Specifically, the global thresholds for shale reservoirs are set to 1×10^{-5} for pressure energy and 0.1 for the attenuation coefficient.

Data Source	Number of Wells and Stages	Sample Size	Formation Coverage
Ordos Basin	4 wells/85 stages	777,703	Mud shale with thin, silty, fine sandstone interbeds
Ordos Basin	7 wells/64 stages	467,836	Medium-grained sandstone

Table 2—Statistical characterization of pressure-derived features in shale and sandstone reservoirs.

Using 5-second analysis windows, a sequence of fracture-pattern labels is obtained, and visualized as a time-varying color map. As detailed in Fig. 10, treating pressure, slurry rate, and proppant concentration are plotted as black, blue, and orange lines, respectively. To visualize how hydraulic energy within the wellbore/formation system is distributed between new fracture-surface creation and dissipation, cumulative pressure fluctuation energy and cumulative attenuation coefficient are shown as purple and green curves. Purple scatter points superimposed on the pressure curve denote identified pressure fluctuation events, serving as proxies for prominent fracture events, with marker size proportional to the intensity of instantaneous stress release. Notably, transient pressure fluctuations are sparse during the first 640 seconds, and the “Fracture interruption” pattern appears frequently. However, the large breakdown-related pressure drop overlaps with the pressure rise induced by rapid rate increase, producing a distorted pressure response in which fracture-induced transient fluctuations are no longer clearly distinguishable. Fracture-pattern diagnosis during near-wellbore breakdown is therefore more susceptible to operational interference, and the results in this interval should be interpreted with caution.

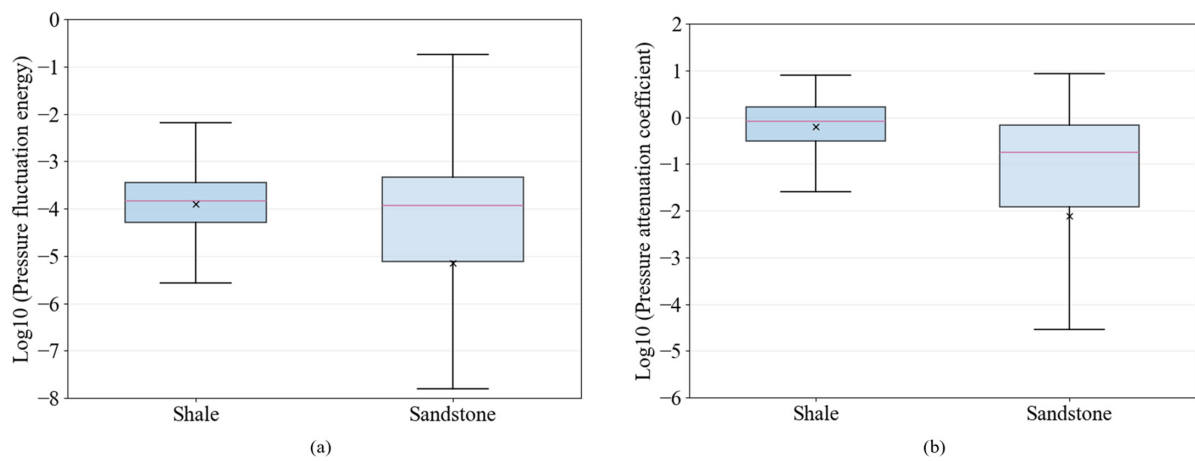


Fig. 9—Boxplot of (a) log10 pressure fluctuation energy and (b) log10 pressure attenuation coefficient for shale and sandstone.

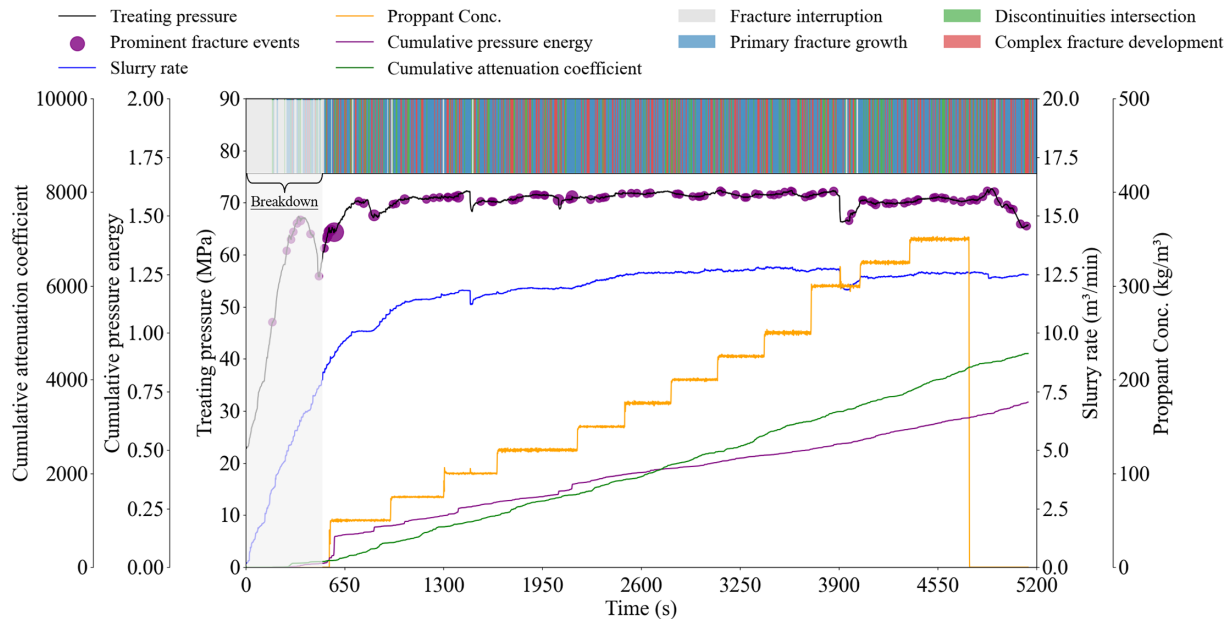


Fig. 10—Fracturing treatment curves and diagnostic results from the proposed method.

Supporting Evidence from Independent Measurements. Supporting Evidence from Microseismic Observations. The MIP-3H well from the Marcellus Shale Energy and Environment Laboratory is selected as Case 1 (Carr et al. 2019). Effective pumping data from 21 stages are analyzed and compared with corresponding microseismic observations. The 1-Hz treating-pressure signal is processed by CWT using the complex Morlet wavelet and 256 linearly discretized frequency scales. Within the 0.3–0.5-Hz diagnostic band, pressure fluctuation energy and attenuation coefficient are calculated, and fracture propagation patterns are classified by the rule-based fuzzy inference system with 5-second windows. Pressure-derived metrics include pressure fluctuation events (as proxies for prominent fracture events), cumulative pressure fluctuation energy, cumulative attenuation coefficient, and fracture-pattern labels, with the temporal proportion of “Complex fracture development” calculated using a 60-second sliding window with a 5-second step. Microseismic metrics include cumulative seismic moment and b -value. Seismic moment scales with the source deformation of individual events and is the fundamental attribute of moment magnitude (Hanks and Kanamori 1979). Cumulative seismic moment captures the overall fracture-energy release and collective fracture activity. The b -value is an important statistical parameter in earthquake monitoring that describes the quantitative distribution of events, commonly written as (Gutenberg and Richter 1941)

$$\log N = a - b \times M, \quad (18)$$

where N is the cumulative microseismic events, M is the magnitude of microseismic events, a is the level of seismic activity, and b is the slope that represents the distribution of microseismic event magnitude. In hydraulic fracturing applications, the b -value is discussed as a potential indicator related to fracture complexity. Higher b -values indicate a larger proportion of small-magnitude events and, in some cases, may be associated with more dispersed fracture growth (Wu et al. 2017). **Fig. 11** summarizes the temporal association between transient-pressure diagnostics and microseismic activity.

Because microseismic monitoring provides discrete events over a broader time scale, its curves are naturally smoother than the pressure-derived metrics. Cumulative pressure fluctuation energy accelerates during intervals of accelerated seismic moment accumulation (e.g., during 2:40–3:05), indicating that more frequent fracture stress-release events excite stronger transient pressure-energy

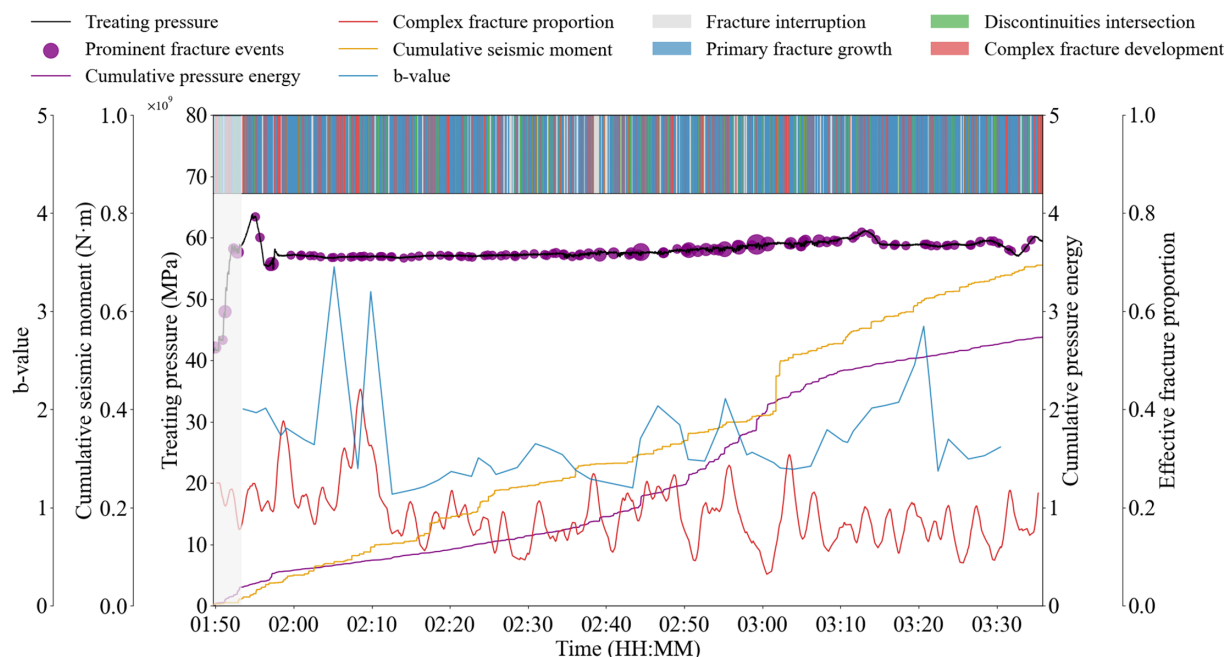


Fig. 11—Pressure diagnostic results from the proposed method, and microseismic observations.

responses. Time windows with elevated b -values moderately correspond to higher proportions of the “Complex fracture development” pattern inferred from pressure analysis, providing moderate support for the interpretation that more distributed and branched fracture paths can induce jointly elevated pressure fluctuation energy and attenuation. Some local discrepancies remain (e.g., during 3:10–3:25), reflecting differences in temporal resolutions and sensing mechanisms of microseismic and pressure measurements. Nevertheless, these moderate temporal associations support the interpretation that the target-band transient pressure fluctuations are more likely induced by fracture stress-state changes than by random perturbations.

To further statistically substantiate this relationship, the analysis is extended to 21 fracturing stages from Case 1 (Fig. 12). Spearman rank correlation shows that the count of microseismic events and cumulative seismic moment are positively correlated with the number of identified pressure fluctuation events and cumulative pressure fluctuation energy, respectively. The overall b -value is positively correlated with cumulative pressure energy associated with the “Complex fracture development” pattern, offering supplementary but non-unique support for the pressure-based interpretation of fracture complexity.

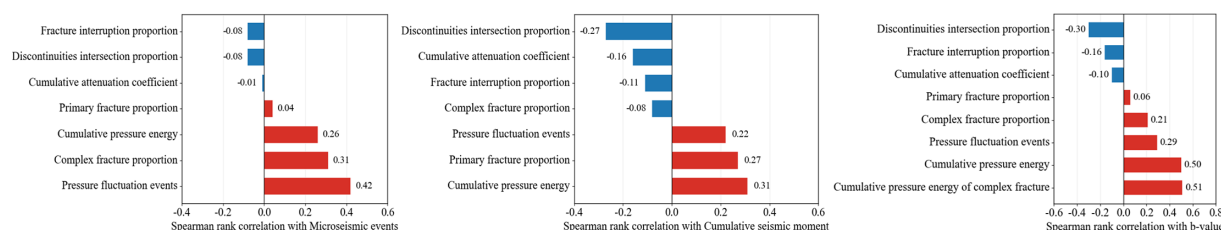


Fig. 12—Spearman rank correlation against microseismic observations.

Supporting Evidence from Distributed Fiber-Optic Observations. A horizontal well in the Junggar Basin is selected as Case 2 to compare the transient-pressure diagnostics with distributed acoustic sensing (DAS) observations. The optical fiber is permanently installed outside the casing of the treatment well, allowing real-time monitoring of the fluid-entry distribution among perforation clusters during stimulation. A total of 35 stages, each with three to six clusters, are included in the analysis. For these stages, the injected fluid volume ranged from 2,004.91 m³ to 2,393.56 m³, the average cluster spacing is 11.5 m, and the mean proppant intensity is 2.0 m³/m. The shale reservoir is influenced by bedding planes and high-stress interbeds, with locally developed natural fractures. This case compares stage-level pressure-derived attributes with DAS-observed cluster fluid-entry uniformity. The pressure-derived attributes include cumulative pressure fluctuation energy and the proportion of the four fracture propagation patterns, while DAS-derived uniformity is quantified by the uniformity index:

$$UI = 1 - \frac{\sigma}{\mu}, \quad (19)$$

where σ is the standard deviation of the fluid-entry volume of clusters per stage, and μ is the average. Spearman rank analysis (Fig. 13a) shows that the uniformity index is positively correlated with cumulative pressure fluctuation energy and the proportions of “Primary fracture growth” and “Complex fracture development,” but negatively correlated with “Discontinuities intersection” and “Fracture interruption.” This suggests that more uniform cluster-level fluid entry is generally associated with higher fracture-fluid efficiency, stronger stress release, and more effective fracture development. However, uneven distribution tends to promote growth barriers controlled by bedding planes or high-stress interbeds, resulting in reduced fluid efficiency and retarded stress accumulation. The effect of pre-existing

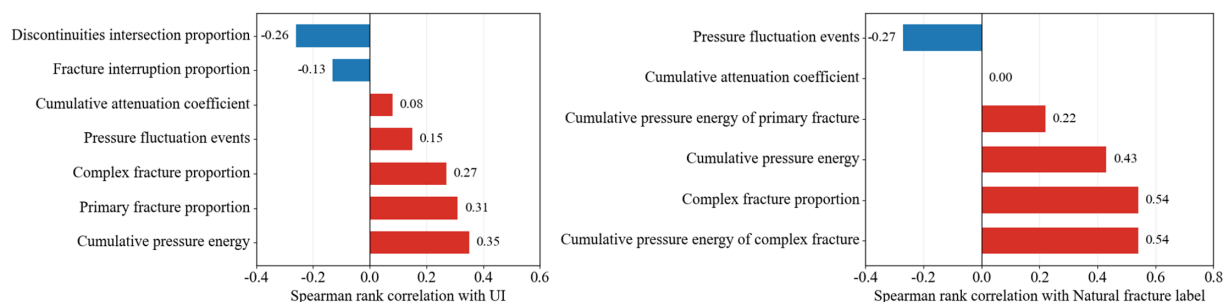


Fig. 13—Spearman rank correlation against (a) cluster fluid-entry uniformity and (b) natural fracture development.

natural fractures is further examined by grouping stages into matrix-dominated intervals (Label 0) and natural-fracture intervals (Label 1). As shown in **Fig. 13b**, cumulative pressure energy associated with “Complex fracture development” exhibits a strongest positive correlation with the natural-fracture development index than that associated with “Primary fracture growth,” providing statistical support that natural fractures may promote fracture deviation, reorientation, and increased complexity.

DAS can also diagnose stage-isolation integrity. **Fig. 14** shows a stage with isolation failure and pronounced interstage communication, where a low-resistance diversion pathway triggers an abrupt pressure drop. After this abnormal pressure drop, the proportion of the “Discontinuities intersection” pattern, shown by the cyan curve, rapidly increases, nearly doubles, and remains high for an extended period, accompanied by decreasing pressure fluctuation energy and increasing attenuation. This divergence suggests that hydraulic energy is dissipated through leakage or diversion rather than converted into new fracture surfaces. At earlier times, pressure fluctuations may still indicate discontinuity-related dissipation even without a distinct DAS interference pattern, because pressure reflects the integrated wellbore/formation response, whereas DAS is more sensitive to pronounced near-wellbore fluid-entry redistribution. Thus, interstage communication can be distinguished from geological discontinuity interaction by the combined signature of a discontinuity-dominated proportion and a sharp abnormal pressure drop.

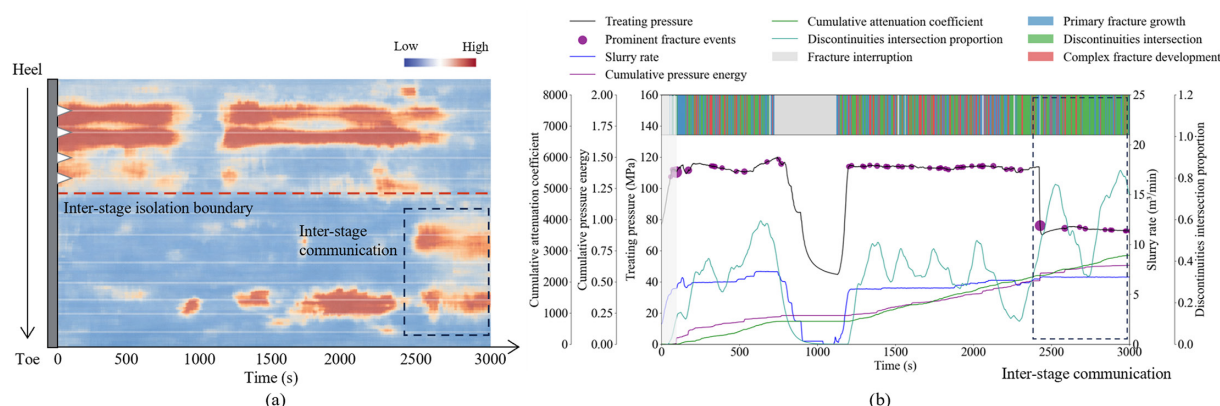


Fig. 14—(a) DAS observation of interstage communication. (b) Diagnostic results of the proposed method for interstage communication.

Supporting Evidence from Tracer Production Data. To provide post-fracturing productivity support, interstage chemical tracer monitoring is analyzed. Chemical tracers quantify the fluid contribution from individual stages during flowback. For 46 shale fracturing stages with comparable oil saturation, Spearman rank correlations are computed between tracer production normalized by stage length and pressure-derived metrics (**Fig. 15**). Tracer production shows a significant positive correlation with cumulative pressure energy attributed to the “Complex fracture development” pattern, suggesting that pressure-inferred fracture complexity corresponds to greater stimulated contact and higher stage contribution. This provides additional support for the productivity relevance of the proposed diagnostics.

Effect of Pressure Acquisition Conditions. Effect of the Pressure Measurement Location. In most field operations, wellhead pressure is readily available, whereas bottomhole pressure requires downhole gauges. Because the proposed attributes are extracted from broadband pressure transients, wellbore transmission can potentially filter or distort broadband responses through frictional losses, fluid compressibility, and wave propagation effects. Therefore, the influence of measurement location on the quantified pressure fluctuation features is evaluated in **Fig. 16**. Overall, both measurements exhibit synchronous temporal evolution in pressure fluctuation energy and attenuation within the diagnostic band, indicating weak sensitivity of the inferred fracture propagation patterns to measurement location. Bottomhole pressure more directly captures local fracture-induced fluctuations because it is inherently decoupled from hydrostatic head and wellbore friction variations. Under the proposed workflow, bottomhole fluctuations produce stronger 0.3–0.5-Hz responses and are less likely to be corrected due to prominent pressure/rate coherence, yielding higher cumulative pressure fluctuation energy and attenuation. By comparison, attenuation is less sensitive to measurement location than energy because location mainly affects the overall signal amplitude while weakly altering the intraband energy distribution. Thus, bottomhole pressure can improve robustness under frequent operational adjustments and reduce the risk of missing effective fracture-growth intervals caused by wellbore transmission effects. Nevertheless, given the limited deployment of downhole gauges, wellhead pressure remains the preferred and more practical input for routine field applications.

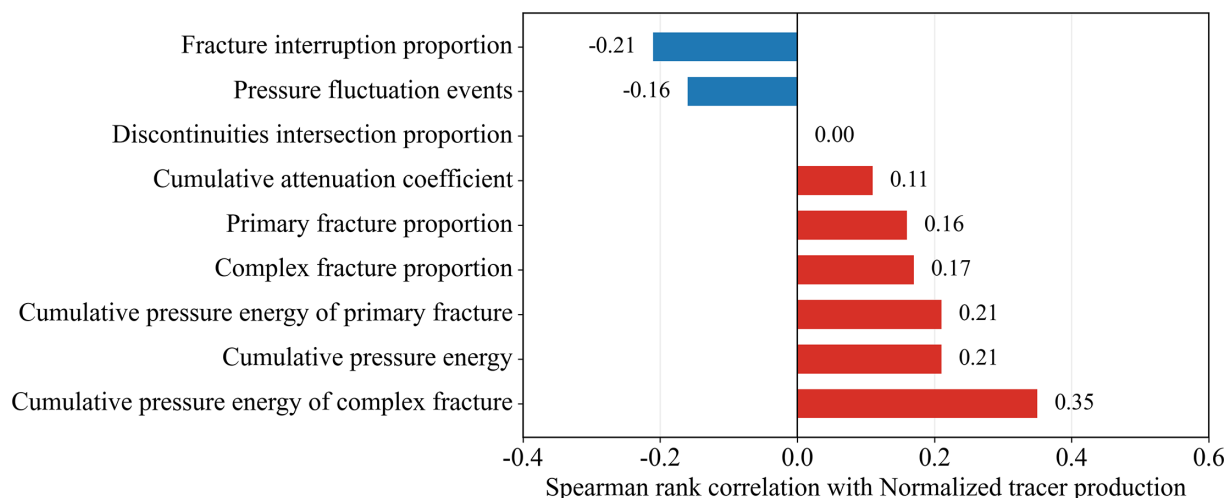


Fig. 15—Spearman rank correlation against stage-level tracer production normalized by stage length.

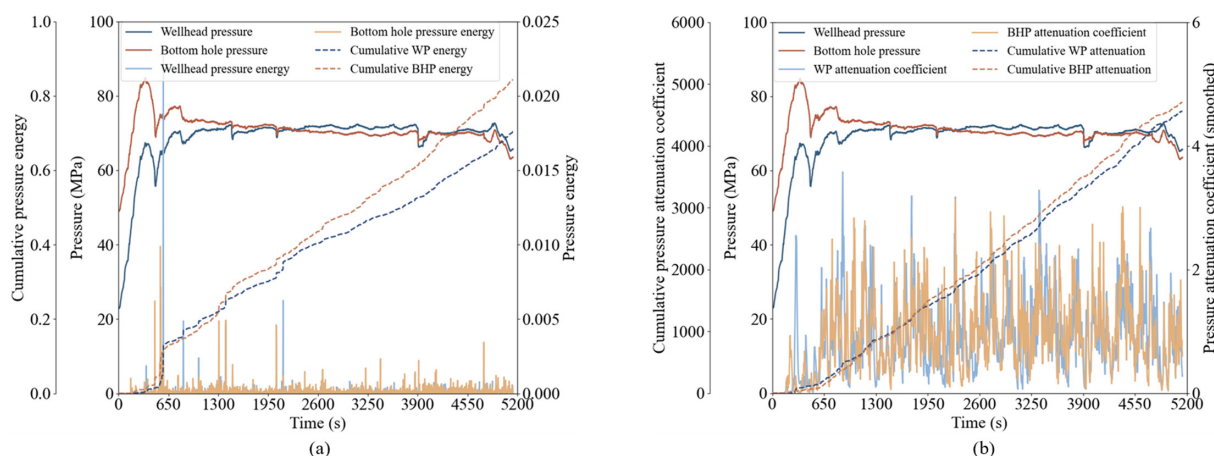


Fig. 16—(a) Comparison of pressure fluctuation energy from wellhead and bottomhole pressure. (b) Comparison of pressure attenuation coefficient from wellhead and bottomhole pressure.

Effect of the Sampling Frequency. Standard 1-Hz field acquisition enables efficient CWT-based fracture propagation identification. Although higher sampling rates help capture broadband impulsive responses and improve the timing accuracy of fracture-event localization, their practical benefit is limited by wellbore resonance, which may dominate high-frequency pressure components and constrain the observable bandwidth of fracture-induced impulses.

Fig. 17 shows the time-frequency energy spectrum obtained from 10-Hz pressure data. The impulse bandwidth extends to approximately 1 Hz but is strongly masked by pronounced wellbore characteristic frequencies at higher frequencies, indicating that the coupled wellbore/fracture system becomes resonance-dominated beyond this range. Thus, increasing the sampling rate does not necessarily expand the usable diagnostic bandwidth. Nevertheless, high-rate acquisition remains advantageous because antialias filtering before downsampling suppresses resonant clutter and prevents its aliasing into the diagnostic band. As shown in Fig. 18, direct 1-Hz acquisition suffers from persistent contamination of wellbore resonance near 0.3 Hz, which may reduce the apparent fracture-induced energy while exaggerating attenuation. In contrast, antialias filtered downsampling from high-rate data mitigates this interference and yields clearer transient signatures for fracture propagation classification.

Discussion

Practical Applicability of the Proposed Method. With this paper, we present a CWT-based diagnostic method that links transient pressure fluctuations to stress release and accumulation of hydraulic fracture propagation. Under standard 1-Hz sampling, pressure fluctuation energy in the 0.4–0.5-Hz sub-band is used to characterize stress release and hydraulic energy conversion into new fracture surfaces, while attenuation toward the 0.3–0.4-Hz sub-band reflects enhanced hydraulic-energy dissipation and hindered stress reaccumulation along the fracture path. Based on these two quantified features, fracture propagation is dynamically classified into four patterns: primary fracture growth, complex fracture development, discontinuities intersection, and fracture interruption.

Strengths of the Method. The physical interpretation of the proposed method is supported by independent monitoring, including microseismic, DAS, and tracer observations. In the Marcellus Shale Energy and Environment Laboratory case, periods of accelerated cumulative seismic moment represent intensified fracture activity and more frequent stress-release events, which excite intense transient pressure drop and accelerated accumulation of pressure fluctuation energy. In addition, intervals with elevated b-values coincide with higher proportions of the “Complex fracture development” pattern, providing moderate support for the interpretation that more complex

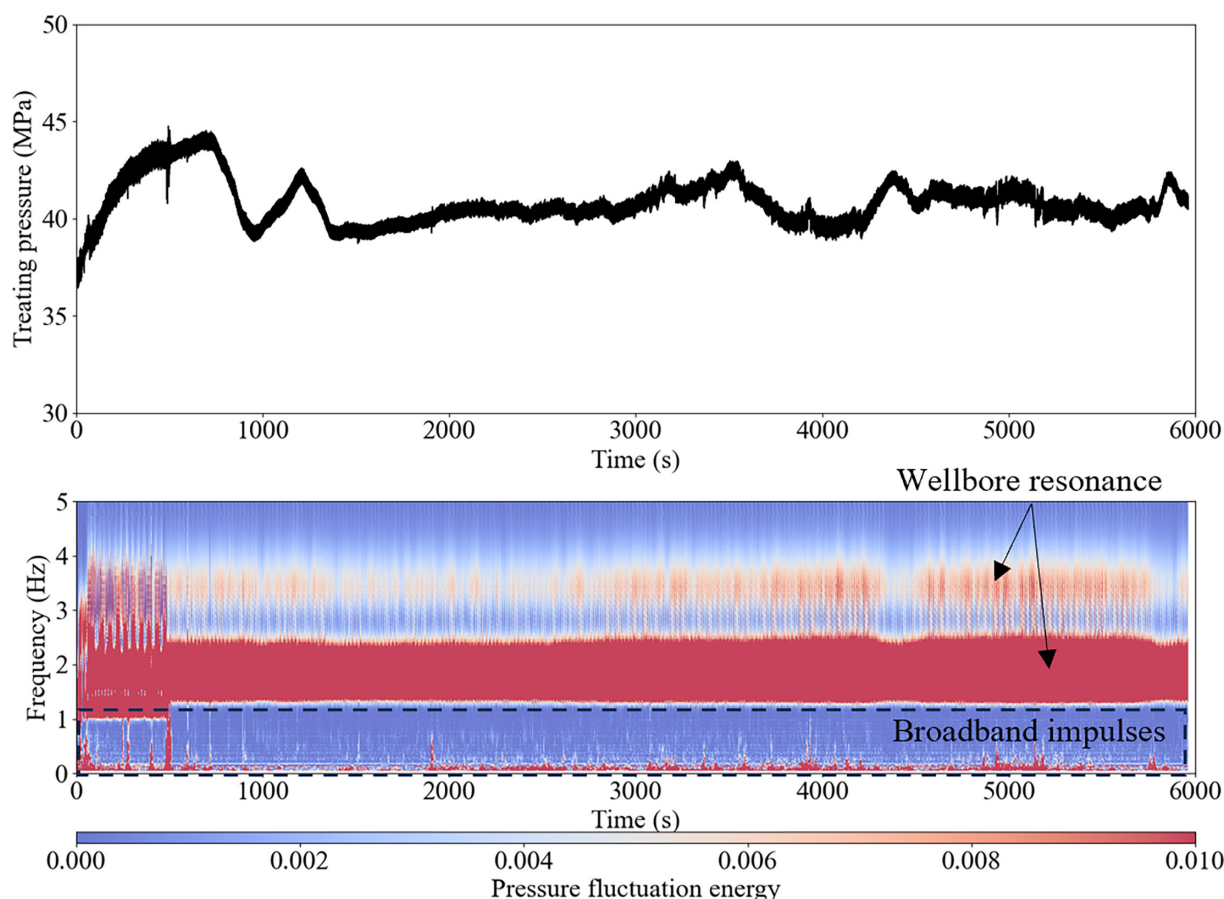


Fig. 17—Time-frequency energy spectra obtained from 10-Hz pressure data.

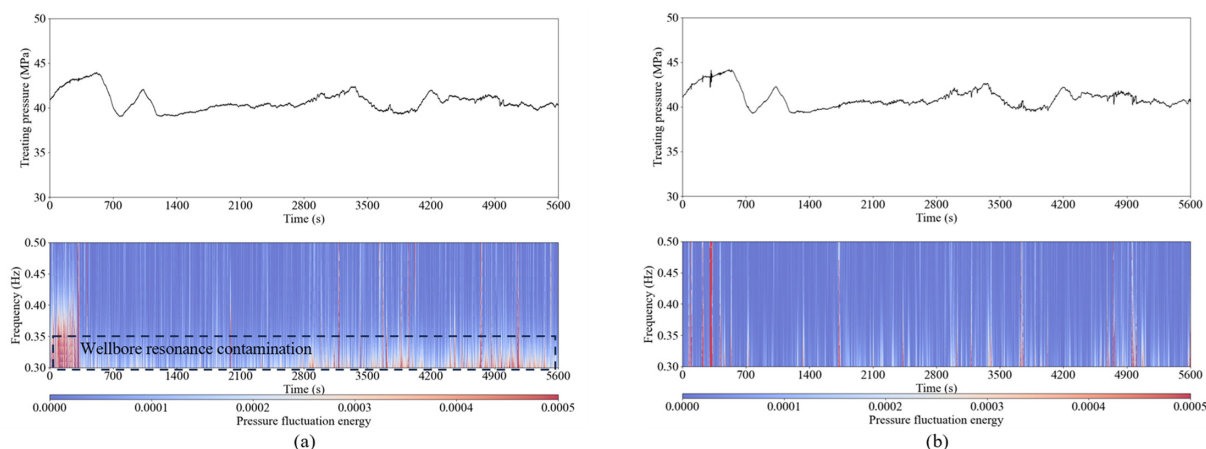


Fig. 18—Time-frequency energy spectra of 1-Hz pressure signals derived from (a) native 1-Hz acquisition and (b) 10-Hz acquisition after antialias filtered downsampling.

fracture paths excite high pressure fluctuation energy and attenuation relatively. In the Junggar Basin case, higher DAS-derived uniformity and dynamic cluster-intake redistribution generally enhance rock breaking and fracture growth, thereby increasing pressure fluctuation energy. Once dominant fractures are established, however, hydraulic energy is preferentially dissipated along these paths, stabilizing pressure and reducing pressure fluctuation energy. This interpretation is further supported by prior numerical simulations of multifracture propagation in horizontal wells (Li et al. 2022b; Wu and Olson 2016). Moreover, the tracer-based productivity comparison supports the view that enhanced fracture-network connectivity improves fracture conductivity and fluid recovery, thereby extending the proposed method from fracture-pattern diagnosis to productivity-relevant stimulation evaluation.

Fundamentally, the method diagnoses whether hydraulic energy in the wellbore/formation system is preferentially converted into new fracture surfaces or dissipated along the fracture path. This enables the identification of fracture-growth intensity, propagation complexity, and abnormal stimulation behaviors such as interstage communication. Previous studies have also shown that transient pressure fluctuations carry diverse fracture-pattern information related to fracture geometry and multifracture competition (Gabry et al. 2024b; Zhang

et al. 2025). Compared with independent monitoring techniques, pressure signals provide a physically grounded and broadly applicable diagnostic source. In practice, the method relies only on routinely recorded treatment data, with wellhead pressure and slurry rate as the minimum inputs, requiring no additional monitoring cost. Independent monitoring data sets are used only as supporting benchmarks rather than deployment prerequisites. By updating fracture-pattern information every few seconds, the method can support real-time treatment optimization. For example, persistently weak “Complex fracture development” accompanied by insufficient pressure energy may indicate poor fracture-fluid efficiency or risks such as interstage communication and screen out, thereby motivating timely adjustments of diversion timing or pumping schedule.

Limitations. Despite these advantages, the proposed method should be regarded as a realistic but simplified system-scale classification of hydraulic-energy allocation rather than a geometric reconstruction of fracture propagation because pressure signals inherently lack spatial information. As a single-point response of the integrated wellbore/formation system, pressure cannot directly resolve fracture geometry, orientation, or stimulation location. Compared with microseismic and DAS measurements, pressure analysis can evaluate the efficiency of hydraulic-energy conversion, cluster-stimulation uniformity, and abnormal interstage communication, but cannot reconstruct 3D fracture geometry or locate stimulation precisely. Spatially different effects, such as asymmetric fracture-wing growth and additional frictional losses along extended fracture paths, may be superimposed in the single-point pressure response, reducing observability and potentially weakening event detection. Therefore, complementary measurements, including offset-well pressure, microseismic, and distributed fiber optic, are valuable for providing spatial and directional constraints for a more comprehensive assessment of fracture geometry, orientation, and complexity.

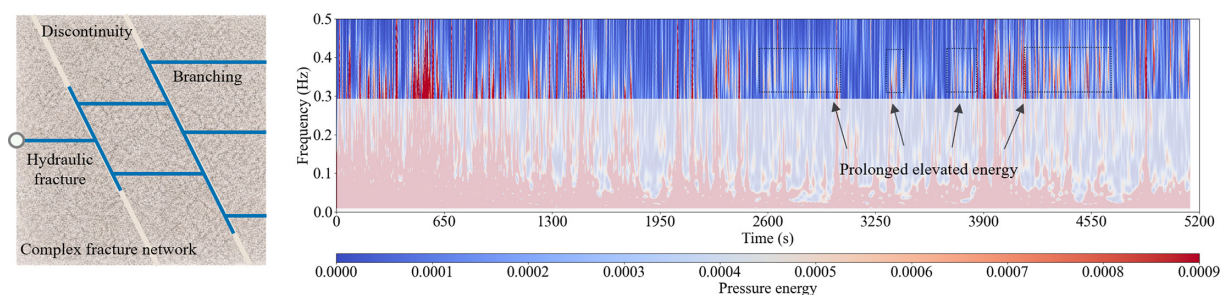
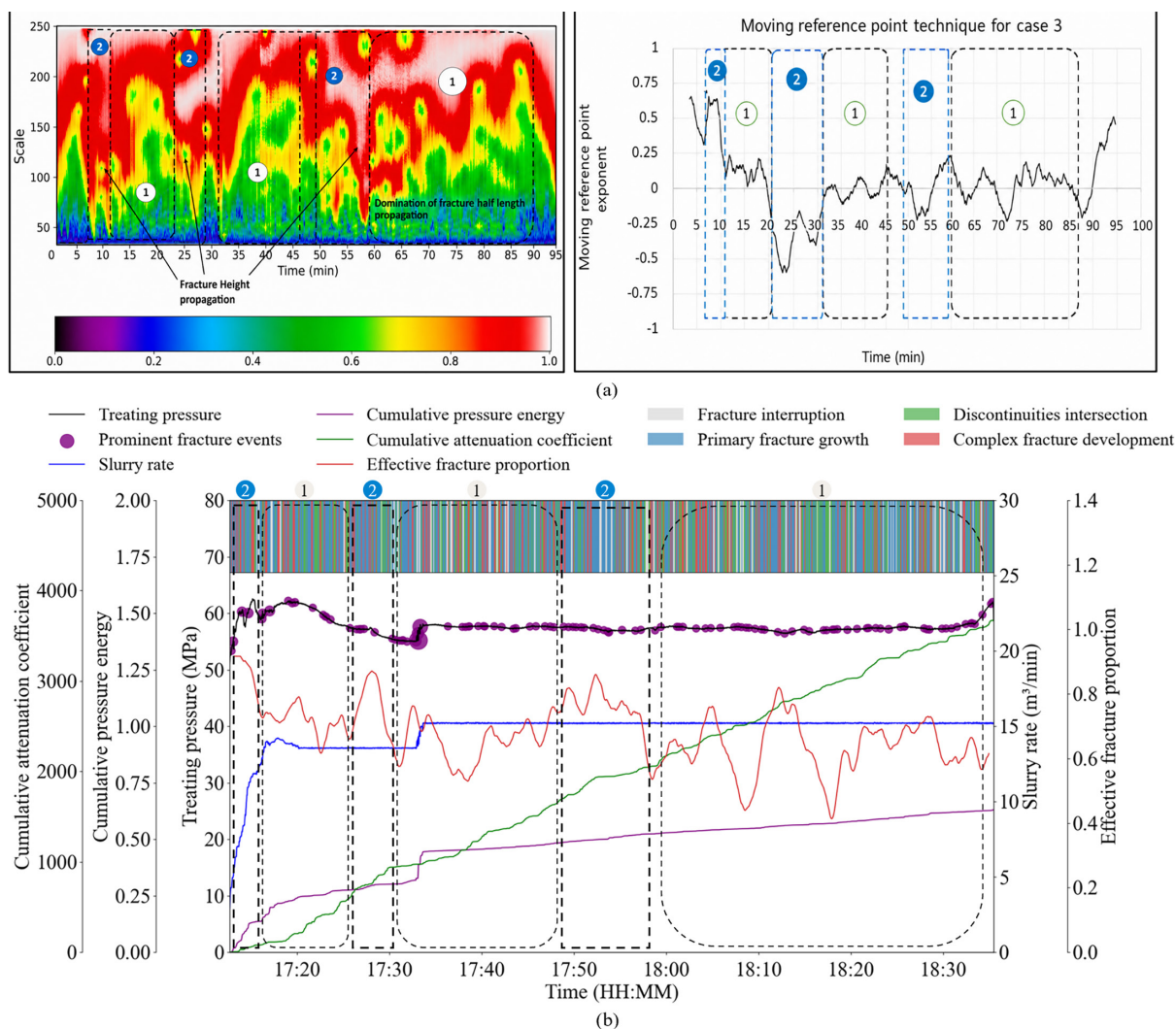
Another limitation is the reduced applicability during near-wellbore breakdown. In this stage, the pressure signal may be dominated by the overall trend, while transient fluctuations are weak, distorted by operational adjustments before the designed pumping rate is reached. In addition, fracture propagation develops the fracture process zone ahead of the fracture tip, where distributed microcracking and material softening reduce the effective cohesion and promote localized failure (Li et al. 2026). By contrast, breakdown is less favorable for generating frequent resolvable transient events because the rock remains largely intact and the threshold for hydraulic energy to be converted into new fracture surfaces is higher. As a result, diagnostic interpretation in this stage becomes less stable. A practical mitigation is to exclude the breakdown stage from the main diagnostic target and focus instead on the more stable fracture-propagation stage. When available, bottomhole pressure and stable hydraulic conditions can further reduce distortion caused by frequent operational adjustments. In addition, wellbore resonance and background noise may affect the time-frequency response. Resonance contaminates specific frequency bands and can be mitigated by antialias filtered downsampling from high-rate data. Background noise reduces the overall signal-to-noise ratio across the time-frequency domain; in such cases, appropriate filtering strategies, such as wavelet denoising, may help to improve the robustness of the interpretation.

Comparison with Existing Pressure-Based Diagnostics. Compared with traditional time-domain pressure analysis, the proposed method better captures discrete fracture-propagation events and complexity. Classical approaches, such as the Nolte-Smith framework, mainly interpret monotonic net-pressure trends to infer fracture geometry evolution (Nolte and Smith 1981), which overlooks discrete transient fluctuations distributed throughout the treatment. Other fluctuation-based studies have related pressure amplitude and frequency to fracture complexity but often remain limited to 1D time-domain interpretation (Bian et al. 2016). In contrast, our method directly quantifies transient pressure fluctuations in the 2D time-frequency domain, avoiding idealized-fracture assumptions and the iterative computations required by moving pressure-reference points (Pirayesh et al. 2015). Combined with CWT multiresolution localization and wavelet-coherence screening, the workflow reduces rate-driven interference and enables time-resolved classification of fracture-propagation patterns. This improves diagnostic resolution and provides a quantitative and interpretable basis for future realization of intelligent fracture pattern diagnosis.

Recent CWT-based diagnostics have demonstrated the value of time-frequency representations for identifying general fracture propagation modes (Gabry et al. 2024b). To provide supporting evidence for the physical consistency of our proposed diagnostic method, we benchmark our results against Case 3 (Stage 28, Well MIP-3H) of Gabry et al. (2024b). In their study, high normalized CWT magnitudes of 0.8–0.95 were validated by MRP analysis and microseismic monitoring as periods dominated by fracture height breakthrough (**Fig. 19**). Our method observes a synchronous increase in effective fracture growth, defined as the sum of primary fracture growth and complex fracture development, during these periods of high normalized CWT values. This agreement supports a shared physical interpretation: The fracture height breakthrough or significant fracture expansion induces strong instantaneous stress release and elevated pressure energy in the time-frequency domain. Building upon these shared benchmarks, the fracture propagation patterns identified by our method can be further subdivided into dominant height growth or length extension. While both methods rely on CWT, they differ in signal decomposition strategy. Gabry et al. (2024b) used a series of linear wavelet scales, which compress high-frequency components to provide a visual interpretation of general fracture modes. However, our workflow prioritizes the most diagnostic high-frequency components (0.3–0.5 Hz) and further enhances the spectral resolution using a linear-frequency binning strategy, thereby expanding the separable structure of high-frequency components that would otherwise be compressed. This refinement not only improves localization of fracture propagation patterns but also facilitates more reliable attenuation analysis of individual fluctuations. Notably, this attenuation analysis extends existing CWT-based approaches, which primarily focus on pressure fluctuation intensity. The proposed method provides a more detailed assessment of fracture complexity and supports real-time tracking of fracture dynamics.

Calibration and Future Work. In the current workflow, thresholds for pressure fluctuation energy and attenuation are derived from statistical distributions as practical global baselines for noise suppression and fuzzy classification. Future work could develop more physically grounded calibration by first establishing formation-specific thresholds from historical stages and then comparing stage-level cumulative pressure fluctuation energy, cumulative attenuation, and inferred fracture-pattern proportions with post-fracture pressure falloff estimates of fracture area, closure behavior, and fracture complexity (Liu et al. 2020; Wei et al. 2024; Zeinabady and Clarkson 2023). Stages showing significant discrepancies could be used for local correction of the feature distributions and global thresholds. Post-fracture pressure falloff analysis provides a cost-effective pathway for post-fracture calibration because it uses routinely acquired pressure data without additional equipment deployment. Combining real-time transient-pressure diagnostics during pumping with post-fracture pressure falloff analysis would enable a unified pressure-based workflow across the full treatment cycle.

In addition, locally prolonged elevated energy is defined as a sustained narrow-band energy concentration near 0.35–0.4 Hz over a finite time interval and likely indicates the development tendency of a complex fracture network (**Fig. 20**). Rapid development of complex fracture paths can trigger successive stress-release episodes and repeated pressure drops, while hydraulic-energy dissipation along tortuous paths delays pressure recovery. As a result, fracture-induced transients occur repeatedly before the system fully re-equilibrates,



enhancing pressure energy over a finite time interval. Meanwhile, relatively stronger high-frequency attenuation shifts and concentrates this sustained elevated energy near 0.35–0.4 Hz, thereby deriving a narrow-band energy response from the broadband impulsive response. This suggests that different fracture behaviors may contain richer diagnostic details in the time-frequency spectrum than can be captured by band-level statistical features alone. In the current workflow, pressure fluctuation energy and attenuation quantify the stress state during fracture propagation, and fuzzy inference provides an interpretable engineering-oriented classifier in data-scarce scenarios. The fuzzy inference results can further serve as preliminary labels for future data-driven interpretation when independent fracture monitoring is insufficient. However, the workflow summarizes each diagnostic band into averaged statistical features and may underutilize fine-scale spectral details at individual frequencies. Future work could enrich the data set and integrate computer-vision learning with time-frequency spectrum to link localized broadband impulses and prolonged energy trails to specific fracture patterns, enabling a more objective and higher-resolution diagnosis of fracture propagation dynamics.

Conclusions

For this paper, we developed a real-time diagnostic method utilizing CWT to extract rate-corrected transient pressure fluctuations associated with hydraulic fracture propagation, enabling the dynamic classification of fracture propagation patterns, including primary fracture growth, complex fracture development, discontinuities intersection, and fracture interruption. The proposed method is supported by independent measurements from microseismic monitoring, distributed fiber-optic measurements, and stage-level tracer production data. It can be concluded that:

1. CWT effectively extracts transient pressure fluctuations intrinsic from the stress release and accumulation of hydraulic fracture propagation, which manifest as broadband impulses in the time-frequency domain. Assisted by wavelet coherence analysis to exclude rate-coupled responses, the 0.3–0.5-Hz frequency components are identified as the optimal diagnostic window for standard 1-Hz sampling data.
2. Pressure fluctuation energy in the 0.4–0.5-Hz sub-band serves as a surrogate for instantaneous stress release of fracture opening and reflects the efficiency with which the hydraulic energy is converted into new fracture surfaces. Pressure fluctuation attenuation toward the 0.3–0.4-Hz sub-band reflects increased hydraulic-energy dissipation along the fracture path and hindered stress reaccumulation. Locally prolonged elevated energy near 0.35–0.4 Hz in the pressure/time-frequency spectrum serves as a potential indicator of enhanced fracture complexity and a diagnosis corroborated by the higher tracer production observed in stages dominated by complex fracture development patterns.
3. The diagnostic precision can be enhanced by mitigating wellbore transmission effects. Bottomhole pressure better preserves local pressure energy by reducing the influence of hydrostatic head and wellbore friction filtering. Furthermore, a higher sampling rate enables antialias filtering that suppresses wellbore resonant contamination within the diagnostic band. This combination effectively isolates fracture-induced responses from wellbore system interference, leading to superior sensitivity to subtle fracture propagation events under frequent operational adjustments.

Acknowledgments

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